

آزمون: بیوشیمی بالینی

زمان شروع آزمون: 12:00

مدت زمان: 25 دقیقه

آزمون مرتبط: بیوشیمی بالینی

تعداد سوالات: 30

تعداد سوالات در هر صفحه: 1

نمره پاسخ صحیح
نمره پاسخ غلط
عدم پاسخ سوال

معمولی	0/000	0/000	1/000	دیابت ایجاد شده در اثر مصرف بیش از حد Diazoxide از کدام نوع محسوب می گردد.	- 1
(1) Type I					
(2) Type II					
(3) ✓ Other Type					
(4) Gestational Diabetes					
معمولی	0/000	0/000	1/000	Hyperosmolar non- ketotic coma بیشتر در کدام نوع دیابت اتفاق می افتد.	- 2
(1) نوع ۱					
(2) ✓ نوع ۲					
(3) دیابت بارداری					
(4) دیابت ناشی از اندوکرینوپاتی					
معمولی	0/000	0/000	1/000	کمای ناشی از هایپراسمولار-هایپر گلیسمی در بیماران دیابتی به خاطر افزایش کدامیک از ترکیبات زیر اتفاق می افتد.	- 3
(1) اسیدهای آمینه					
(2) ✓ گلوکز					
(3) کتون بادی ها					
(4) اسیدهای چرب					
صفحه 1 از 10					
192.168.14.6 Power By www.faradidsoft.com					

1. Diabetes caused by excessive use of Diazoxide is classified as which type?

1. Type I
2. Type II
3. Other Type ✓
4. Gestational Diabetes

2. Hyperosmolar non-ketotic coma occurs more frequently in which type of diabetes?

1. Type 1
2. Type 2 ✓
3. Gestational diabetes
4. Diabetes caused by endocrinopathy

3. Hyperosmolar-hyperglycemic coma in diabetic patients occurs due to an increase in which of the following compounds?

1. Amino acids
2. Glucose ✓
3. Ketone bodies
4. Fatty acids

4. What is the main cause of hypoglycemia in newborns with galactosemia?

1. Accumulation of galactose-1-phosphate ✓
2. Inhibition of glucose-6-phosphatase by galactose-1-phosphate
3. Accumulation of galactose in the blood
4. Inhibition of aldolase B

5. The oxidative stress produced in certain organs of galactosemia patients and diabetic patients is related to which of the following?

1. Excessive consumption of NADH,H⁺ by dehydrogenase
2. Excessive consumption of NADPH,H⁺ by aldose reductase ✓
3. Formation of AGE compounds
4. Decreased vitamin D level

6. Hyperuricemia caused by excessive fructose consumption in beverages is due to a decrease in which compound?

1. Leptin
2. Calcium
3. Triglyceride
4. Phosphate ✓

7. In McArdle disease, which of the following symptoms is present?

1. Myoglobinuria ✓
2. Metabolic alkalosis
3. Hepatomegaly
4. Anemia

8. In SLOS (Smith-Lemli-Opitz Syndrome), which intermediate of cholesterol biosynthesis accumulates?

1. Mevalonate
2. Isopentenyl pyrophosphate
3. Lanosterol ✓
4. 7-dehydrocholesterol

9. In SLOS and mevalonic aciduria patients, what are the hematologic findings?

1. Thrombocytopenia – increased RBC
2. Thrombocytopenia – leukocytosis ✓
3. Thrombocytosis – leukocytosis
4. Anemia – leukopenia

10. The inherited deficiency of which of the following compounds leads to Hyperchylomicronemia (Type I Fredrickson disease)?

1. LPL ✓
2. ACAT
3. LDL receptor
4. Apo A1

11. In Fredrickson Type I disease, which of the following clinical complications does NOT occur?

1. Increased triglycerides
2. Pancreatitis
3. Hepatosplenomegaly
4. Tendon xanthoma ✓

12. For early detection of glomerular injury, which laboratory test is better than the others?

1. Random urine creatinine
2. Microalbumin-to-creatinine ratio in random urine ✓
3. Microalbuminuria
4. Serum creatinine

13. Which of the following best reflects renal tubular injury?

1. Urine osmolality / Plasma osmolality ✓
2. Microalbuminuria
3. Proteinuria
4. UmicroAlb / UCr

14. Which of the drugs used to lower blood pressure causes hyperkalemia?

1. Triamterene ✓
2. Acetazolamide
3. Furosemide
4. Thiazide

15. An increase in urea without an increase in creatinine is possible in which of the following conditions?

1. Protein malnutrition
2. Urinary tract tumors
3. Hemolytic anemia ✓
4. Diarrhea and vomiting

16. In alcoholic hepatitis, which statement about ALT and AST enzymes is correct?

1. The ALT-to-AST ratio is greater than 1
2. ALT and AST enzymes may rise above 1000 IU/L
3. The rise in ALT and AST resembles the pattern caused by hepatitis A virus
4. The ALT-to-AST ratio remains less than 1 and their values stay below 300 ✓

17. In the autoimmune biliary tract diseases PBC and PSC, which lead to cholestasis, which marker shows a significant increase?

1. AST
2. ALT
3. ALP ✓
4. Albumin

18. Which of the intermediates of cholesterol biosynthesis is required as a membrane anchor for proteins?

1. Geraniol
2. Isopentenyl
3. Farnesyl ✓
4. Lanosterol

19. In liver cirrhosis, all of the following changes are observed EXCEPT:

1. Increased conjugated bilirubin ✓
2. Increased PT
3. Increased PTT
4. Decreased serum albumin

20. In the autoimmune biliary tract diseases PBC and PSC, which lead to cholestasis, which marker shows a significant increase?

1. Placental (fetal)
2. Intestinal
3. Hepatic (liver) ✓
4. Bone

21. In Tangier disease, which of the following has the inherited defect that leads to early cardiac problems?

1. ABCA1 ✓
2. ABCG5
3. ABCG8
4. LCAT

22. Which of the following liver enzymes is NOT GPI-anchored?

1. GGT
2. ALP
3. SGPT (ALT) ✓
4. 5'-nucleotidase

23. In the formation of nascent HDL, which of the following apolipoproteins plays the essential role?

1. Apo C-II
2. Apo A-V
3. Apo E
4. Apo A-I ✓

24. Which of the following symptoms does NOT occur in phenylketonuria (PKU)?

1. Liver cirrhosis ✓
2. Self-injurious behavior (self-harm)
3. Paralysis
4. Mousy odor of urine and sweat

25. In which of the disorders of the phenylalanine catabolism pathway does Ochronosis occur?

1. Tyrosinemia III
2. Tyrosinemia II
3. Alkaptonuria ✓
4. PKU

26. Which of the following disorders leads to skin-eye (oculocutaneous) complications, and whose eye complications require differentiation from those caused by herpes simplex virus in the differential diagnosis?

1. Hartnup Disease
2. Tyrosinemia II ✓
3. MSUD
4. Celiac disease

27. Brain damage in MSUD (Maple Syrup Urine Disease) results from the accumulation of which of the following?

1. Ammonia ✓
2. GABA
3. Decreased alpha-keto acids of branched-chain amino acids
4. Increased propionyl-CoA

28. In which of the following conditions does a caramel or burnt-sugar (maple syrup) odor of urine caused by the presence of sotolon NOT occur?

1. Tyrosinemia type III ✓
2. Pregnancy
3. Prolonged starvation
4. MSUD

29. The biosynthesis of tetrahydrobiopterin (BH₄) takes place from which of the following compounds?

1. ATP
2. GTP ✓
3. Phenylalanine
4. Oxaloacetate

30. In the oxidative decarboxylation of branched-chain amino acids, which of the following vitamins is required?

1. Thiamine ✓
2. Folate
3. Cobalamin (vitamin B12)
4. Pyridoxine